be a stable, hospitable one: a safe place in which to grow old. The discovery of the Big Bang and the expansion of the universe, however, ruled that out. Our universe is changing, and we've only just begun to develop the theories and observations to understand exactly how. The developments of the last few years, and even months, are finally allowing us to paint a picture of the far future of the cosmos.

I want to share that picture with you. The best measurements we have are only consistent with a handful of final apocalyptic scenarios, some of which may be confirmed or ruled out by observations we're making right now. Exploring these possibilities gives us a glimpse of the workings of science at the cutting edge, and allows us to see humanity in a new context. One which, in my opinion, can bring a kind of joy even in the face of total destruction. We are a species poised between an awareness of our ultimate insignificance and an ability to reach far beyond our mundane lives, into the void, to solve the most fundamental mysteries of the cosmos.

To adapt a line from Tolstoy, every happy universe is the same; every unhappy universe is unhappy in its own way. In this book, I describe how small tweaks to our current, incomplete knowledge of the cosmos can result in vastly different paths into the future, from a universe that collapses on itself, to one that rips itself apart, to one that succumbs by degrees to an inescapable expanding bubble of doom. While we explore the evolution of our modern understanding of the universe and its ultimate end, and grapple with what that means *for us*, we'll encounter some of the most important concepts in physics and see how these connect not just to cosmic apocalypses,[fn3] but also to the physics of our everyday lives.

QUANTIFYING COSMIC DOOM

Of course, for some of us, cosmic apocalypses are already a daily concern.

I remember vividly the moment I found out that the universe might end at any second. I was sitting on Professor Phinney's living room floor with the rest of my undergraduate astronomy class for our weekly dessert night, while the professor sat on a chair with his three-year-old daughter on his lap. He explained that the sudden space-stretching expansion of the early universe, cosmic inflation, was still such a mystery that we don't have any idea why it started or why it ended, and we have no way of saying that it won't happen again, right now. No assurance existed to tell us that a rapid,